

Oumar Tirera

2nd Year B.Tech Student | Computer Science & Engineering (AI & ML) | Sharda University

📞 +91 8787338826 | ✉ oumartambatirera@gmail.com | 🔗 www.linkedin.com/in/oumar-tirera-5a7a7b16b

PROFESSIONAL SUMMARY

Motivated 2nd year B.Tech student specializing in Computer Science (AI & ML) at Sharda University, with hands-on experience in web development and a strong foundation in machine learning tools and frameworks. Passionate about solving real-world problems through data-driven approaches and aspiring to grow as an ML Engineer.

SKILLS

- Languages: Python, C++, JavaScript, SQL
- Web Technologies: HTML5, CSS3, React.js, TypeScript
- Machine Learning: NumPy, Pandas, Scikit-learn, Matplotlib (Data Visualization)
- Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebooks
- Soft Skills: Problem-Solving, Time Management

CERTIFICATIONS & TRAINING

Full Stack Web Development – Udemy | May 2024 – July 2024

- Completed an intensive 70+ hour bootcamp covering both frontend and backend web development.
- Built responsive web applications using React.js and TypeScript, ensuring cross-platform compatibility.
- Implemented RESTful APIs using Node.js and integrated PostgreSQL databases for dynamic content management.
- **Capstone Project:** Developed a fully functional e-commerce platform (Gamers Zone) with user authentication and CRUD operations, alongside a personal portfolio website.

PROJECTS

Self-Optimizing Network Infrastructure Using Quantum-Assisted Routing

Tools: Python, NetworkX, D-Wave Ocean SDK (Simulated Annealing), Matplotlib, SDN Principles

- Developed a hybrid quantum-classical framework to address NP-hard traffic engineering problems by offloading routing optimization from a classical SDN controller to a quantum-assisted processing environment.
- Conducted a literature review of 50+ research papers to identify performance gaps in classical heuristics.
- Designed a network simulation in Python and implemented a QUBO (Quadratic Unconstrained Binary Optimization) model to achieve globally optimal routing paths using a simulated quantum annealer.

AI Travel Agent — Full-Stack AI-Powered Trip Planning Application

Tools: Python, FastAPI, LangChain (ReAct), Next.js 14, TypeScript, PostgreSQL, Redis, Docker, Gemini API, Ollama

- Built a full-stack AI travel planning app featuring a LangChain ReAct agent that orchestrates 6 custom tools (flight search, hotel search, restaurant finder, budget calculator, itinerary builder, web search) to generate personalized trip itineraries with real-time pricing data.
- Integrated multiple LLM backends (local Ollama models: Qwen3:8b, Gemma2:9b, and cloud Google Gemini 2.0 Flash) with automatic model routing, enabling a local-first, privacy-conscious architecture with cloud fallback.
- Implemented SSE (Server-Sent Events) streaming for real-time AI responses, Redis-based session persistence and API result caching (30–60 min TTLs), and rate limiting — reducing redundant API calls and improving UX.
- Containerized the entire 4-service stack (Next.js, FastAPI, PostgreSQL, Redis) using Docker Compose for one-command deployment; built a responsive Next.js frontend with dark/light theme, model selector, and PDF export of itineraries.

EDUCATION

Sharda University — Greater Noida, India

B.Tech in Computer Science and Engineering (AI & ML) | Aug 2024 – Present (Expected 2028)